Electric Vehicle Market Performance & Insights

Project Report · Ujjawal Rathod · Individual Project · 29 October 2026

1. Introduction

1.1 Project Overview

The global electric vehicle (EV) market is large, fragmented and changing quickly. Decision makers need to know which vehicle categories sell best, who the leading manufacturers are, which battery technologies are adopted, how far vehicles can drive, and how satisfied customers are. This project analyses a market dataset of **300,000 records and 37 columns** (ev_market_train.csv with 200,000 rows and ev_market_test.csv with 100,000 rows, combined into one data source). It covers 15 vehicle categories, 200 manufacturers, 7 market regions, 10 manufacturing countries and model years 2018–2026.

The solution is an interactive **Tableau dashboard of 9 visualizations (V1–V9)** and a **4-point Tableau Story**. Both are published on Tableau Public and embedded in a **Python Flask web application deployed on Vercel**. The work is organised around three stakeholder scenarios: Emily Johnson (Market Strategy Director), David Chen (Automotive BI Manager) and Dr. Sarah Williams (Sustainable Mobility Researcher).

1.2 Purpose

The purpose of the project is to turn a large, raw EV market dataset into clear, interactive insights that answer each stakeholder's questions in seconds. Specifically, it aims to:

- Identify the vehicle categories and model years that drive sales volume.
- Benchmark manufacturers and measure how fragmented the market is.
- Show battery technology adoption and how it relates to driving range.
- Compare customer satisfaction across categories.
- Show production by country and sales by region to support expansion decisions.

2. Ideation Phase

2.1 Problem Statement

PS	I am	I'm trying to	But
PS-1	Emily Johnson, Market Strategy Director responsible for the EV product portfolio	Identify top-selling EV categories, track multi-year sales trends (2018–2026) and align vehicle range with customer satisfaction	Extracting clear answers from a 300,000-row spreadsheet takes too long, so it is hard to see which categories truly lead in volume
PS-2	David Chen, Automotive BI Manager tracking competitor share, production hubs and regional demand	Benchmark the Top 10 manufacturers, analyse production across 10 countries and spot regional expansion opportunities	Standard spreadsheets and basic filters hide true market share and product demand across regions
PS-3	Dr. Sarah Williams, Sustainable Mobility Researcher evaluating battery chemistries and range	Evaluate how battery technologies (NMC, LFP, NCA, Sodium-Ion, Solid-State) relate to driving range, efficiency and customer ratings over time	It is hard to separate the effect of vehicle body type from the effect of battery chemistry across model years

2.2 Empathy Map Canvas

The primary empathy map is for Emily Johnson, with summary maps for David and Sarah.

- **Says:** "Which EV categories are driving our highest annual sales?" and "We need to identify high-growth segments across 2018–2026 before allocating budget."
- **Thinks:** Investing in a segment without verifying demand could waste millions; with 200 manufacturers, guessing trends without data is too risky.
- **Does:** Compares categories, manufacturers, range and ratings side by side; uses filters (region, battery type, model year) to test strategic scenarios.
- **Feels:** Overwhelmed by raw CSV files with 37 columns; pressured by deadlines; confident and relieved when an interactive dashboard answers her questions quickly.

David focuses on market share, manufacturer ranking and geography. Sarah focuses on whether newer battery chemistry really improves range and satisfaction.

2.3 Brainstorming

Five ideas were listed and prioritised:

1. Merge the train and test files into one 300,000-row dataset and build a 9-visual Tableau dashboard (feasibility high, importance high).
2. Use Level of Detail (FIXED) calculations for Top 10 manufacturer market share so filtering does not inflate percentages to 100%, and start rating and line axes at zero to avoid visual distortion.
3. Build three persona mini-dashboards and combine them into a 4-point Tableau Story (feasibility high, importance high).
4. Build a Flask web application that embeds the dashboard and story, and deploy it on Vercel.
5. Train a regression model to predict future resale value and battery degradation. This was judged out of scope for this project and is listed under Future Scope.

3. Requirement Analysis

3.1 Customer Journey Map

The journey was mapped for all three personas across four stages:

1. **Raw data exploration:** the user faces large CSV files and unclear fields.
2. **KPI and visual discovery (V1–V9):** the user finds category, manufacturer, battery, range, rating and geography insights.
3. **Interactive dashboard cross-filtering:** the user filters by region, model year, battery type and category, and clicks charts to filter others.
4. **Executive storytelling and web portal access:** the user follows the 4-point story and opens the Vercel web app to share findings.

3.2 Solution Requirement

Functional requirements

- Combine the two CSV files into one data source of 300,000 rows.
- Provide 9 visualizations covering sales, manufacturers, trends, battery technology, range, geography and ratings.
- Provide dashboard filters for Market Region, Model Year, Battery Type and Category, plus click-to-filter actions.

- Calculate manufacturer market share with a FIXED Level of Detail expression.
- Provide a 4-point story, one overview and one point per persona.
- Publish the dashboard and story online and embed them in a web page.

Non-functional requirements

- Usability: clear titles, consistent colours and filters in a fixed location.
- Performance: filters and cross-filtering respond in about 1–2 seconds.
- Accuracy: null values documented, market share relative to the full market, axes anchored at zero where needed.
- Accessibility: the dashboard is available through a public link and a responsive web page.

3.3 Data Flow Diagram

Data flows in five steps:

1. The raw files ev_market_train.csv (200,000 rows) and ev_market_test.csv (100,000 rows) enter as the external data source.
2. **Data ingestion and union:** the files are combined into 300,000 rows × 37 columns and field types and geographic roles are assigned.
3. **Data store:** the combined data is held as a Tableau extract.
4. **BI analytics:** the FIXED market share is computed and visuals V1–V9, the dashboard and the 4-point story are built.
5. **Web delivery:** the published Tableau Public links are embedded in a Flask application on Vercel, where stakeholders view insights and apply filters.

(Insert the data flow diagram image from the Requirement Analysis folder here.)

3.4 Technology Stack

Layer	Technology
Data source	CSV files (train and test), Tableau Union connector
Data storage	Tableau extract (.twbx)
Calculations	Tableau calculated fields and FIXED Level of Detail expressions

Layer	Technology
Visualization	Tableau Desktop and Tableau Public (dashboard and story)
Web application	Python Flask, HTML and CSS
Deployment and tools	Vercel, GitHub, Visual Studio Code

4. Project Design

4.1 Problem Solution Fit

The problem is that stakeholders cannot quickly extract reliable answers from a very large, unstructured dataset. The solution fits because one interactive dashboard answers all three personas' questions, keeps market share correct under filtering, and is accessible through a single web link. The fit with the customer is strongest where the pain is highest: time lost in spreadsheets and unreliable shares when data is filtered.

(Add the Problem-Solution Fit canvas from your Project Design folder here.)

4.2 Proposed Solution

- **Dashboard:** 9 linked visualizations with 4 global filters and click-to-filter actions.
- **Accurate market share:** Market Share % = SUM(Annual Sales Units) / MIN({ FIXED : SUM(Annual Sales Units) }), so each manufacturer is compared with all 200 manufacturers even when the Top 10 filter is on.
- **Story:** a 4-point narrative (overview plus Emily, David and Sarah), each point with a key-insight caption.
- **Web portal:** a Flask application that embeds the live dashboard and story and is hosted on Vercel.

4.3 Solution Architecture

The architecture has three layers: (1) data layer, the two CSV files unioned into a Tableau extract; (2) analytics layer, calculated fields, FIXED LOD logic, 9 worksheets, the dashboard and the story published to Tableau Public; (3) presentation layer, the Flask application on Vercel that embeds the published dashboard and story for stakeholders.

(Add the solution architecture diagram from your Project Design folder here.)

5. Project Planning & Scheduling

5.1 Project Planning

The project was executed in sequential phases:

1. Ideation: problem statements, empathy maps and brainstorming.
2. Requirement analysis: customer journey map, solution requirements, data flow diagram and technology stack.
3. Project design: problem-solution fit, proposed solution and solution architecture.
4. Project planning: sprint and task planning.
5. Project development: data preparation, 9 visualizations, dashboard, story and web integration.
6. Performance testing.
7. Documentation and final submission.

(Paste the sprint table and dates from your Project Planning Template here.)

6. Functional and Performance Testing

6.1 Performance Testing

Parameter	Result
Data rendered	300,000 rows × 37 columns from the union of the train and test files
Data preprocessing	Field types checked; model_year set to discrete; manufacturing_country given a Country/Region geographic role; about 1% of annual_sales_units and 2% of customer_rating values are null and were kept because Tableau ignores nulls in aggregates; 1 duplicate vehicle_id found and retained as negligible
Filters	Market Region, Model Year, Battery Type and Category on all worksheets; Top 10 Manufacturer (V2); Top 5 Category (V3); click-to-filter from V1, V4 and V7

Parameter	Result
Calculated fields	Total Sales; Sales (Millions); Market Share % (FIXED LOD); quick table calculations for percent of total and percent difference
Dashboard design	9 visualizations (V1–V9)
Story design	4 story points covering all 9 visualizations

7. Results

7.1 Output Screenshots

(Insert a screenshot for each item below.)

- **V1 Annual sales by category:** Two-Wheeler Scooter leads with roughly 36–38 million units per region, followed by Sedan, Hatchback and Compact SUV.
- **V2 Top 10 manufacturers and market share:** the market is highly fragmented; the top manufacturer, Ampere Works, holds only about 0.6% of total sales.
- **V3 Sales trend by model year:** sales by category across 2018–2026.
- **V4 Battery type share:** NMC 33.7%, LFP 32.0%, NCA 17.3%, Sodium-Ion 9.9% and Solid-State about 7%.
- **V5 Average range by category:** range depends on vehicle type, from about 550 km for the Luxury Sedan to about 105 km for the Two-Wheeler Scooter.
- **V6 Range by battery type over years:** average range stays flat at about 324 km across chemistries from 2018 to 2026.
- **V7 Manufacturing country map:** China leads production (about 237 million units), followed by India and Vietnam, across 10 countries.
- **V8 Customer rating by category:** satisfaction is consistent at about 4.08–4.09 out of 5 in every category.
- **V9 Region × category heatmap:** sales are spread evenly across the 7 market regions (about 2,323.3 million units in total), so product mix matters more than region.
- **Final dashboard, 4-point story and Flask web app:** add full-page screenshots.

8. Advantages & Disadvantages

Advantages

- One interactive view replaces manual analysis of 300,000 rows.
- The FIXED LOD calculation keeps market share accurate when filters are applied.
- Persona-based story makes insights easy to present to executives.
- Publicly accessible through Tableau Public and a Vercel web app, with no software needed by viewers.

Disadvantages

- Tableau Public data is publicly visible, so it is unsuitable for confidential data.
- The analysis is descriptive; it does not predict future sales or prices.
- Missing values (about 1–2% in some fields) are excluded from averages and totals.
- The web app depends on the availability of Tableau Public embeds.

9. Conclusion

The project converted a 300,000-row EV dataset into a clear, interactive decision tool. The analysis shows a fragmented market (no manufacturer above about 0.6% share), strong sales in Two-Wheeler Scooters and Sedans, a mix of battery chemistries led by NMC and LFP, and driving range that depends on vehicle category rather than battery chemistry. Customer satisfaction is consistently high at about 4.1 out of 5. The dashboard, story and web application give all three stakeholders quick, reliable answers.

10. Future Scope

- Add predictive models for resale value and battery degradation, using the resale and cycle-life fields.
- Add forecasting of sales by category and region beyond 2026.
- Connect to a live data source or cloud data warehouse so the dashboard refreshes automatically.
- Add price and efficiency analysis and a manufacturer comparison tool.
- Add user logins and role-based views for different stakeholders.

11. Appendix

Source Code

Flask application files: app.py, requirements.txt, templates/index.html and static/style.css (see the GitHub repository below).

Dataset Link

ev_market_train.csv (200,000 rows) and ev_market_test.csv (100,000 rows): [PASTE DATASET LINK HERE]

GitHub & Project Demo Link

- GitHub repository: https://github.com/ujjawalrathod404/Electric-Vehicle-market-performace-insights
- Tableau Public dashboard and story: https://public.tableau.com/app/profile/ujjawal.rathod4051/viz/ElectricVehicleMarketPerformanceInsights_17931389708610/EVMarketStory
- Flask web app on Vercel: https://electric-vehicle-market-performance.vercel.app/